

CENTRALISED TRAIL AUDIT COMPLIANCE ENGINE

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Application Logs → Parse → Detect PII → Mask PII → Store Securely → Generate Hash → Auditor checks/export

Example:

Original:

Before Masking:

Credit Card = 4532-1234-5678-9012

After masking:

Credit Card = ****_****_****-9012

What these System actually do ?

- Generally companies / Apps will generate audit logs .
- This system collects audit logs from companies / User Application.
- Parse the audit logs
- It detects the personal information PII(like security number , Credit card number)
- It masks the personal information.
- It stores PII securely .
- Each time it generates hash SHA-256 for every audit logs.
- Compliance officer will check and filter the audit logs .
- It provides security , protection and reliability

Use-Case Flow: Submit and Process Audit Log

Use Case ID

UC-01

Use Case Name

Submit and Process Audit Log

Primary Actor

Application / Enterprise System

Supporting System

Centralized Audit Trail Compliance Engine

Goal

To receive an enterprise audit log, validate it, mask sensitive PII, generate an integrity hash, and securely store the audit record.

Preconditions

1. The audit logging system is available.
2. The enterprise application is authorized to submit audit logs.
3. The audit log is submitted in JSON format.

Main Success Scenario

1. The Enterprise Application submits a structured JSON audit log to the compliance engine.
2. The system receives the audit log.
3. The system validates the JSON structure and required fields.
4. The system identifies sensitive PII fields such as credit card numbers and Social Security Numbers.
5. The system masks all detected PII fields.
6. The system generates a SHA-256 integrity hash for the masked audit record.
7. The system stores the masked audit record and its integrity hash in secure storage.
8. The system returns a successful processing response to the Enterprise Application.
9. The use case ends successfully.

Alternate Flow

A1. Invalid Audit Log

1. At Step 3, the system detects that the submitted JSON is malformed or missing required fields.
2. The system rejects the audit log.
3. The system records the validation failure.
4. The system returns an error message to the Enterprise Application.
5. The invalid record is not stored as a valid audit record.
6. The use case ends.

A2. PII Detection Failure

1. At Step 4, the system cannot reliably process the submitted PII fields.

2. The system rejects the audit log instead of storing potentially sensitive cleartext data.
3. The system records the processing failure.
4. The system notifies the submitting application.
5. The use case ends unsuccessfully.

Postconditions**Success**

The audit record is stored with PII masked and a valid SHA-256 integrity hash.

Failure

The invalid or unsafe audit log is not stored as a valid compliance record.